

Symbolic Composition Hashing: An Algebraic Sponge over Finite Fields with ZK-Friendly Structure

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ABSTRACT. We introduce *Symbolic Composition Hashing (SCH)*, a new algebraic framework for constructing cryptographic hash functions over finite fields based on the symbolic composition of triangular and affine polynomial maps. The construction defines a broad family of permutation-based designs whose security is governed by the algebraic complexity of the resulting multivariate systems. To capture this complexity, we formalize the *Symbolic Composition Inversion Problem (SCIP)*, a hardness assumption describing the difficulty of inverting or colliding composed polynomial permutations when only their symbolic representation is available. Under SCIP and standard sponge-based arguments, SCH provides a general foundation for preimage and collision resistance. We analyze structural properties of the construction, including algebraic degree growth, diffusion behavior, and resistance to generic algebraic attacks, and we support these observations with preliminary experimental evaluations. Rather than proposing a finalized hash standard, this work establishes a foundational algebraic family and hardness framework intended to support further cryptanalytic study and future cryptographic design.

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1. Introduction

Cryptographic hash functions are a fundamental component of modern security systems, underpinning applications ranging from data integrity and authentication to blockchain protocols and privacy-preserving constructions. While most standardized hash designs rely on bit-oriented primitives, there has been growing interest in algebraic and multivariate approaches that operate natively over finite fields, motivated by their structural richness and potential resistance to emerging attack models.

Existing algebraic hash constructions typically focus on specific instantiations or fixed designs, leaving open the question of whether broader algebraic *frameworks* can be defined in which security properties emerge from general structural principles rather than ad hoc choices. In particular, the role of symbolic composition of polynomial maps as a source of algebraic hardness remains underexplored.

In this work, we propose *Symbolic Composition Hashing (SCH)*, a general construction paradigm based on iterated compositions of triangular and affine polynomial maps over finite fields. By treating the composed permutation symbolically rather than numerically, SCH induces dense multivariate systems whose inversion and collision properties depend on the algebraic complexity of the composition itself.

To reason about the security of this family, we formalize the *Symbolic Composition Inversion Problem (SCIP)*, which captures the difficulty of recovering preimages or

collisions given only the symbolic representation of a composed polynomial permutation. SCIP is intended as a unifying abstraction for the algebraic hardness arising from symbolic composition and serves as a focal point for future cryptanalytic investigation.

We emphasize that this work does not claim a finalized or standardized hash primitive. Instead, our goal is to define a coherent algebraic family, articulate the underlying hardness assumptions, and analyze the resulting structural properties. Security arguments are therefore conditional and heuristic, relying on SCIP and generic sponge-based reasoning, with the expectation that further cryptanalysis will refine and challenge these assumptions.

Contributions. This paper makes the following contributions:

- We introduce *Symbolic Composition Hashing (SCH)*, a general algebraic framework for hash construction based on symbolic polynomial composition.
- We formalize the *Symbolic Composition Inversion Problem (SCIP)* as a hardness assumption governing preimage and collision resistance in this setting.
- We analyze structural properties of SCH, including algebraic degree growth, diffusion behavior, and resistance to generic algebraic attacks.
- We provide preliminary experimental results illustrating degree growth and diffusion characteristics across multiple composition depths.
- We identify open problems and research directions for further cryptanalysis and parameter selection.

2. Related Work

Traditional hash functions. Cryptographic hash function design has been dominated by bit-oriented constructions. The Merkle–Damgård paradigm [7] and the sponge construction underlying SHA-3/Keccak [6] process data through Boolean round functions optimised for diffusion and nonlinearity. These designs are extremely fast in software and hardware, but operate at the bit level and yield no algebraic structure exploitable in proof systems.

Arithmetization-oriented hashes. The growth of zero-knowledge proof systems and verifiable computation has motivated hash functions defined natively over large prime fields. MiMC [3] pioneered this direction with a univariate power map iterated over many rounds; its low multiplicative complexity produces small arithmetic circuits but limits algebraic degree growth per round. The HADES design strategy [16] introduced a mix of full and partial S-box layers, yielding Poseidon [1] and its successor Poseidon2 [17], which remain the most widely deployed arithmetization-friendly hashes. Rescue [2] alternates high- and low-degree power maps for efficient verification in both R1CS and AIR representations. Griffin [4] combines a Feistel-like structure with low-degree nonlinearities. More recent designs include Anemoi [14], which introduces the “flystel” open butterfly structure, Reinforced Concrete [15], which mixes field and decomposition-based nonlinearities, and Tip5 [18], which targets recursive STARKs. Structural commonality. Despite their variety, the constructions above share a common pattern: a fixed nonlinear round function is iterated a prescribed number of times, and security is argued by analysing algebraic degree growth, differential propagation, and resistance to Gröbner basis or interpolation attacks. Degree growth is typically driven by a single monomial power map (x^d for small d) applied coordinate-wise, with diffusion provided by an MDS or near-MDS linear layer.

The SCH approach. The present work departs from this pattern. Instead of a power-map S-box, SCH uses triangular polynomial maps—composed with affine layers—to build a permutation whose each component is a dense multivariate polynomial. Because triangular maps contribute degree growth *multiplicatively* across rounds (Theorem 5.2), the algebraic degree of the full permutation reaches d_{tri}^k after k rounds, yielding rapid growth even for moderate parameters. The hardness assumption (SCIP, Definition 7.1) is therefore a multivariate polynomial inversion problem rather than a discrete-log or power-map assumption, connecting SCH to the MQ problem studied in multivariate cryptography [5].

Algebraic attacks. Security against algebraic attacks is bounded by the complexity of Gröbner basis computation on the resulting polynomial system. Faugère’s F4 [11] and F5 [12] algorithms, together with the XL family [10], are the most efficient known solvers. Their complexity is governed by the degree of regularity D_{reg} , for which Bardet, Faugère, and Salvy [13] provide asymptotic estimates under the semi-regularity assumption. We use these bounds to derive concrete security estimates (Section 7) and validate them experimentally on reduced instances (Section 9.12).

3. Preliminaries

3.1. Finite Fields and Polynomial Maps. Let p be a prime and let $\mathbb{F}_p$ denote the finite field of cardinality p . For a positive integer n , we consider vectors $x = (x_1, \dots, x_n) \in \mathbb{F}_p^n$ and polynomial maps $F : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ of the form

$$F(x) = (f_1(x), \dots, f_n(x)),$$

where each coordinate function $f_i \in \mathbb{F}_p[x_1, \dots, x_n]$ is a multivariate polynomial.

Polynomial maps over finite fields provide a natural algebraic setting for cryptographic constructions, as their structural properties—such as degree growth, variable interaction, and compositional behavior—can be analyzed using tools from computational algebra while remaining compatible with finite-field arithmetic.

3.2. Affine and Triangular Polynomial Maps. A polynomial map $T : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is called *triangular* (in the *lower-triangular* convention) if each coordinate function depends only on the current and *preceding* variables, i.e.,

$$T(x_1, \dots, x_n) = (t_1(x_1), t_2(x_1, x_2), \dots, t_n(x_1, \dots, x_n)),$$

with $t_i \in \mathbb{F}_p[x_1, \dots, x_i]$. A particularly useful bijective subclass is the *unitriangular* form

$$T(x_1, \dots, x_n) = (x_1 + c, x_2 + g_2(x_1), \dots, x_n + g_n(x_1, \dots, x_{n-1})),$$

where $c \in \mathbb{F}_p$ and each $g_i \in \mathbb{F}_p[x_1, \dots, x_{i-1}]$. Such maps are bijective and admit efficient inversion by *forward substitution*. Moreover, their Jacobian is lower-triangular with all diagonal entries equal to 1, hence $\det(J_T) = 1$ over $\mathbb{F}_p$.

3.3. Composition and Algebraic Degree. Given polynomial maps $F, G : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$, their composition is defined as

$$(F \circ G)(x) = F(G(x)).$$

Repeated composition of polynomial maps generally increases algebraic complexity, as reflected in the growth of algebraic degree and the densification of variable interactions.

The *algebraic degree* of a polynomial map F , denoted $\deg(F)$, is defined as

$$\deg(F) = \max_{1 \leq i \leq n} \deg(f_i),$$

where $\deg(f_i)$ is the total degree of the polynomial f_i . Under composition, degrees may grow rapidly, especially when nonlinear triangular maps are interleaved with affine transformations.

Degree growth and diffusion under composition are central indicators of resistance to algebraic and interpolation-based attacks, and they play a key role in the security analysis of algebraic cryptographic constructions.

3.4. Symbolic Representation of Polynomial Maps. In this work, we distinguish between the *symbolic* and *numerical* representations of polynomial maps. A symbolic representation treats each coordinate polynomial as an explicit algebraic expression in $\mathbb{F}_p[x_1, \dots, x_n]$, preserving full structural information about variable dependencies and coefficients.

Symbolic representations arise naturally when polynomial maps are constructed through explicit algebraic composition. While such representations enable formal analysis of structure and degree growth, they also induce dense multivariate systems whose inversion or collision-finding may be computationally difficult.

The distinction between symbolic and numerical representations is fundamental to the construction introduced in this paper, as the hardness of inverting symbolically composed maps forms the basis of the security assumptions underlying Symbolic Composition Hashing.

3.5. Cryptographic Hashing and Sponge Constructions. A cryptographic hash function maps inputs of arbitrary length to fixed-length outputs while aiming to provide properties such as preimage resistance, second-preimage resistance, and collision resistance. Sponge constructions offer a flexible framework for building hash functions from an underlying permutation, absorbing input blocks into an internal state and squeezing out output blocks.

In permutation-based hash designs, the security of the hash function depends critically on the structural properties of the internal permutation, including its diffusion behavior and resistance to generic attacks. In the context of algebraic constructions, these properties are closely tied to algebraic degree growth and the complexity of the induced polynomial systems.

The SCH framework adopts the sponge paradigm while focusing on permutations generated through symbolic composition of polynomial maps, thereby linking algebraic structure directly to cryptographic security considerations.

4. Symbolic Composition Hashing Construction

This section defines the *Symbolic Composition Hashing (SCH)* construction at an abstract level. The goal is to specify a general family of algebraic permutations over finite fields whose security derives from symbolic composition rather than from a fixed numerical design. Concrete parameter choices and instantiations are deferred to Section 5.

4.1. Construction Overview. Let p be a prime and let $\mathbb{F}_p$ denote the finite field of cardinality p . Let $n \geq 1$ be the state dimension. SCH constructs a permutation

$$P : \mathbb{F}_p^n \longrightarrow \mathbb{F}_p^n$$

as a symbolic composition of affine and triangular polynomial maps.

The defining feature of SCH is that the permutation P is treated as an *explicit symbolic polynomial mapping*, obtained by algebraic composition, rather than as a black-box numerical routine. This symbolic representation induces dense multivariate polynomial systems whose inversion and collision properties form the basis of the security analysis.

4.2. Affine and Triangular Components. SCH is built from two classes of bijective polynomial maps:

Affine maps. An affine map $A : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is defined as

$$A(x) = Mx + b,$$

where $M \in \mathbb{F}_p^{n \times n}$ is invertible and $b \in \mathbb{F}_p^n$. Affine maps provide global linear mixing of variables and preserve bijectivity.

Triangular maps. A triangular map $T : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ (lower-triangular convention) is of the form

$$T(x_1, \dots, x_n) = (x_1 + c, x_2 + g_2(x_1), x_3 + g_3(x_1, x_2), \dots, x_n + g_n(x_1, \dots, x_{n-1})),$$

where $c \in \mathbb{F}_p$ and each g_i is a polynomial over $\mathbb{F}_p$ in the indicated variables. Such maps are bijective and admit efficient inversion by forward substitution. In addition, J_T is lower-triangular with ones on the diagonal, so $\det(J_T) = 1$.

4.3. Symbolic Composition. Let $k \geq 1$ be an integer referred to as the *composition depth* or *number of rounds*. The SCH permutation P is defined as the composition

$$P = A_k \circ T_k \circ A_{k-1} \circ T_{k-1} \circ \dots \circ A_1 \circ T_1,$$

where each A_i is an affine map and each T_i is a triangular map.

The permutation P is constructed *symbolically* by explicit algebraic substitution. That is, the coordinate polynomials of P are obtained by expanding the composition in $\mathbb{F}_p[x_1, \dots, x_n]$, rather than by treating each round as an opaque numerical operation.

As the composition depth k increases, the resulting coordinate polynomials exhibit increasing algebraic degree, variable interaction, and symbolic density. These properties are central to the intended resistance against algebraic and interpolation-based attacks.

4.4. Design Rationale. The SCH construction is guided by the following principles:

- **Bijectivity.** Each component map is invertible, ensuring that the composed mapping P is a permutation on $\mathbb{F}_p^n$.
- **Degree Growth.** Nonlinear triangular layers introduce controlled algebraic degree growth, which is amplified by interleaving affine mixing layers.
- **Symbolic Density.** Explicit symbolic composition produces dense multivariate polynomials, discouraging sparse or low-degree representations exploitable by algebraic solvers.
- **Structural Flexibility.** Parameters such as the state dimension n , composition depth k , and polynomial degrees are left adjustable within the framework.

4.5. From Permutation to Hash Function. The SCH construction defines a family of algebraic permutations. To obtain a cryptographic hash function, the permutation P is embedded into a sponge construction, as specified in Section 5. Standard sponge arguments then relate the security of the hash function to the difficulty of inverting or colliding the underlying permutation.

4.6. Relation to SCIP. The security of SCH is formalized through the *Symbolic Composition Inversion Problem (SCIP)*. Informally, SCIP captures the difficulty of recovering preimages or collisions given only the symbolic polynomial representation of P , without access to its round decomposition.

By defining P through symbolic composition rather than numerical design, SCH aims to maximize the algebraic complexity encountered by generic inversion and collision-finding algorithms. A formal statement of SCIP and its role in security analysis is given in Section 7.

5. Specification

This section specifies a *reference instantiation* of the Symbolic Composition Hashing (SCH) framework for the purposes of reproducibility, experimental evaluation, and implementation clarity. The parameter choices, constant-generation procedures, encoding rules, and mode semantics defined here are not claimed to be optimal or finalized. Rather, they provide a concrete baseline instance of the general SCH construction introduced in Section 4, intended to support empirical analysis and third-party cryptanalytic investigation.

5.1. Parameters. Let p be a publicly specified prime and let $\mathbb{F}_p$ denote the field $\mathbb{F}_p$. The sponge state has size $n = r + c$, where r is the *rate* and c is the *capacity*. Parameters are selected so that generic sponge bounds provide 2^{128} or 2^{256} security against generic collision and preimage attacks:

$$p^{c/2} \geq 2^{128}, \quad p^c \geq 2^{256}, \quad p^r \geq 2^{256}.$$

The first inequality targets collision resistance via the birthday bound, the second targets preimage resistance derived from the capacity, and the third ensures that each absorbed rate block spans a space of size at least 2^{256} when encoding inputs as field elements (see Section 5.2).

These inequalities are conservative sizing targets used for parameter selection; Table 1 lists representative instantiations intended to meet these bounds under standard sponge-security heuristics rather than exact tight proofs.

Primality of p is assumed as part of parameter selection; primality testing is not part of the SCH protocol.

TABLE 1. SCH parameter sets as implemented in the reference code.

Name	Prime field p	$\log_2(p)$	n	r	c	k	d_{tri}
toy31 (Sec. 6)	31	≈ 5	3	2	1	3	3
sch128	$2^{127} - 1$	127	12	8	4	7	3
sch192	$2^{191} - 1$	191	16	10	6	9	3
sch256	$2^{255} - 19$	255	20	12	8	11	4

Notes. State size is $n = r + c$; k denotes the number of composition rounds and d_{tri} the maximum degree of each unitriangular layer. Under standard sponge heuristics, generic collision work is $\approx p^{c/2}$ and generic preimage work is $\approx p^c$ (capacity-driven). For **sch128** with $\log_2 p = 127$ and $c = 4$, generic collision work is $\approx 2^{254}$, well above the 128-bit target. Higher parameter sets provide proportionally larger margins. The **toy31** set is for illustration only and provides no cryptographic security.

5.2. Encoding and Padding. SCH hashes arbitrary byte strings by (i) padding the input to a multiple of a fixed byte block size, (ii) packing each block into an element of $\mathbb{F}_p$, and (iii) absorbing those field elements into the sponge rate. This specification is deterministic, injective, and unambiguous.

Byte order and notation. Let $m \in \{0, 1\}^{8\ell}$ be a message consisting of ℓ bytes. Write $m = (m_0, \dots, m_{\ell-1})$, where each $m_i \in \{0, \dots, 255\}$ denotes a byte. Packing within each block follows a *little-endian* convention: the first byte of a block is the least significant.

Block size. Let

$$B = \lfloor \log_{256}(p - 1) \rfloor = \left\lfloor \frac{\log_2(p - 1)}{8} \right\rfloor.$$

Then $256^B \leq p - 1$, ensuring that every packed B -byte block corresponds to a unique element of $\mathbb{F}_p$ without modular wraparound. For example, for $p = 2^{255} - 19$, we obtain $B = 31$.

Padding (10* padding). Let $\text{pad}(m)$ be the padded byte string obtained by appending a single byte **0x80**, followed by zero bytes until the total length is a multiple of B . Formally, define

$$m' = m \parallel \text{0x80} \parallel \underbrace{\text{0x00} \parallel \dots \parallel \text{0x00}}_{t \text{ bytes}},$$

where t is the unique integer in $\{0, \dots, B - 1\}$ such that

$$|m'| \equiv 0 \pmod{B}.$$

This padding rule is injective and prevents ambiguity for messages with trailing zero bytes.

Packing bytes into field elements. Partition m' into L consecutive blocks of B bytes:

$$m' = \text{blk}_0 \parallel \text{blk}_1 \parallel \dots \parallel \text{blk}_{L-1}, \quad |\text{blk}_j| = B.$$

Define the packing map $\text{pack} : \{0, \dots, 255\}^B \rightarrow \mathbb{F}_p$ by

$$\text{pack}(\text{blk}_j) = \sum_{i=0}^{B-1} \text{blk}_j[i] \cdot 256^i \in \{0, \dots, 256^B - 1\} \subset \{0, \dots, p - 2\} \subset \mathbb{F}_p.$$

Let

$$\alpha_j = \text{pack}(\text{blk}_j) \in \mathbb{F}_p, \quad 0 \leq j \leq L - 1.$$

Rate absorption mapping. Let the sponge state be $s \in \mathbb{F}_p^n$, where $n = r + c$ and the first r coordinates constitute the rate. Packed elements are absorbed in blocks of r : for each block $(\alpha_{jr}, \dots, \alpha_{jr+r-1})$, update

$$s_i \leftarrow s_i + \alpha_{jr+i-1} \pmod{p}, \quad i = 1, \dots, r,$$

followed by application of the permutation P . If the final block contains fewer than r elements, the remaining rate coordinates are left unchanged before applying P .

Domain separation. To prevent cross-protocol collisions (e.g., between hashing and PRF-style usage), domain separation is achieved by initializing the state with a public tag $\tau \in \mathbb{F}_p$:

$$s \leftarrow (\tau, 0, 0, \dots, 0) \in \mathbb{F}_p^n.$$

The tag τ is derived deterministically from an ASCII identifier (e.g., "SCH" or "SCH-Hash") using the same packing rule, truncating to B bytes if necessary. All experiments in this paper use $\tau = \text{pack}(\text{"SCH"})$.

Squeezing. After absorption, output is produced by reading up to r rate coordinates and applying P between output blocks, as in the standard sponge construction. For a t -field-element digest $(z_1, \dots, z_t)$: extract $\min(r, t - \text{output so far})$ elements from $(s_1, \dots, s_r)$, then apply $s \leftarrow P(s)$ if more elements are needed, and repeat until all t elements have been produced. If a byte-string digest is required (e.g., 32 bytes), each z_i is serialized in little-endian base-256 using exactly B bytes and truncated to the desired output length.

Implementation note. The encoding above avoids modular reduction during packing (since $256^B < p$), is constant-time with respect to the message, and is directly compatible with field-native zero-knowledge circuits.

5.3. Permutation Definition. The internal permutation $P : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is defined as the composition of k rounds, each consisting of a nonlinear triangular map followed by an affine mixing map. This ordering promotes rapid algebraic degree growth through nonlinear layers, while affine layers ensure global mixing of variables across the state.

Formally, the i -th round is defined as

$$f_i = A_i \circ T_i,$$

where the components are specified as follows.

- **Triangular layer.** $T_i : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is a *lower-triangular unitriangular* polynomial map of the form

$$T_i(x_1, \dots, x_n) = (x_1 + c_i, x_2 + g_{i,2}(x_1), x_3 + g_{i,3}(x_1, x_2), \dots, x_n + g_{i,n}(x_1, \dots, x_{n-1})),$$

where $c_i \in \mathbb{F}_p$ is a round constant and each $g_{i,j} \in \mathbb{F}_p[x_1, \dots, x_{j-1}]$ is a polynomial of bounded degree. This form is bijective and admits efficient inversion by *forward substitution*. Moreover, the Jacobian matrix J_{T_i} is lower-triangular with all diagonal entries equal to 1, and therefore

$$\det(J_{T_i}) = 1 \quad \text{over } \mathbb{F}_p.$$

- **Affine mixing layer.** $A_i : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is an affine map defined by

$$A_i(x) = M_i x + b_i,$$

where $b_i \in \mathbb{F}_p^n$ and $M_i \in \text{SL}_n(\mathbb{F}_p)$ satisfies $\det(M_i) = 1$. The affine layer provides linear diffusion across all state coordinates.

The restriction $\det(M_i) = 1$ simplifies symbolic Jacobian analysis and ensures volume-preserving affine mixing; it is not required for bijectivity and may be relaxed in alternative instantiations of the framework.

The full permutation is given by the composition

$$P = f_k \circ f_{k-1} \circ \cdots \circ f_1 = A_k \circ T_k \circ \cdots \circ A_1 \circ T_1.$$

Since each triangular layer T_i has unit Jacobian determinant and each affine layer A_i satisfies $\det(M_i) = 1$, the composed permutation P is bijective with

$$\det(J_P) = 1.$$

Without access to the internal round decomposition, inverting P corresponds to solving a dense multivariate polynomial system whose algebraic degree and symbolic density grow rapidly as a function of the number of rounds k .

5.4. Structural Properties. We now state the main structural results that underpin the security analysis.

Theorem 5.1 (Bijectivity and unit Jacobian determinant). *Let $P = A_k \circ T_k \circ \cdots \circ A_1 \circ T_1$ where each $T_i : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ is a unitriangular polynomial map and each $A_i(x) = M_i x + b_i$ with $M_i \in \text{SL}_n(\mathbb{F}_p)$. Then P is bijective with $\det(J_P) = 1$ over $\mathbb{F}_p$.*

Proof. Each T_i is bijective: writing

$$T_i(x_0, \dots, x_{n-1}) = (x_0 + c_i, x_1 + g_{i,1}(x_0), \dots, x_{n-1} + g_{i,n-1}(x_0, \dots, x_{n-2})),$$

the inverse is recovered uniquely by forward substitution, starting from $x_0 = y_0 - c_i$ and then setting $x_j = y_j - g_{i,j}(x_0, \dots, x_{j-1})$ for $1 \leq j \leq n-1$. Each affine layer $A_i(x) = M_i x + b_i$ is bijective because $M_i \in \text{SL}_n(\mathbb{F}_p)$ is invertible, so $A_i^{-1}(y) = M_i^{-1}(y - b_i)$. Hence P is bijective as a composition of bijections.

For the Jacobian determinant, each T_i is unitriangular, so its Jacobian J_{T_i} is lower-triangular with all diagonal entries equal to 1; hence $\det(J_{T_i}) = 1$. Each A_i has Jacobian $J_{A_i} = M_i$ with $\det(M_i) = 1$ by hypothesis. By the chain rule for polynomial maps,

$$J_P = J_{A_k} \cdot J_{T_k} \cdot J_{A_{k-1}} \cdot J_{T_{k-1}} \cdots J_{A_1} \cdot J_{T_1},$$

$$\text{so } \det(J_P) = \prod_{i=1}^k \det(M_i) \cdot \det(J_{T_i}) = 1. \quad \square$$

Theorem 5.2 (Degree growth bound). *Let each unitriangular map T_i satisfy*

$$\max_{1 \leq j \leq n} \deg(g_{i,j}) \leq d_{\text{tri}}.$$

Then the total degree of the composed permutation satisfies

$$\deg(P) \leq d_{\text{tri}}^k.$$

Proof. We proceed by induction on the number of rounds k .

Base case ($k = 1$). After one round, $P = A_1 \circ T_1$. Each coordinate of T_1 has degree at most d_{tri} , and A_1 is affine (degree 1), so $\deg(A_1 \circ T_1) \leq d_{\text{tri}} = d_{\text{tri}}^1$.

Inductive step. Suppose that after $k-1$ rounds the composed map has degree at most d_{tri}^{k-1} . One additional round applies T_k followed by A_k . Since T_k is unitriangular of degree d_{tri} , composing T_k with a map of degree D yields degree at most $D \cdot d_{\text{tri}}$ (each variable in $g_{k,j}$ is replaced by a polynomial of degree $\leq D$, and $g_{k,j}$ has degree $\leq d_{\text{tri}}$). Applying A_k (degree 1) preserves this bound. Hence $\deg(P_k) \leq d_{\text{tri}}^{k-1} \cdot d_{\text{tri}} = d_{\text{tri}}^k$. $\square$

Proposition 5.1 (Monomial density growth). *Let P_k denote the composed permutation after k rounds, and let N_k be the number of monomials with non-zero coefficients in an arbitrary coordinate polynomial of P_k . For generic choices of round constants, N_k is non-decreasing in k , and empirically grows at a rate close to $\binom{n+d_{\text{tri}}^k}{n}$ (the total number of monomials of degree $\leq d_{\text{tri}}^k$ in n variables).*

This proposition is supported experimentally by the degree-tracking experiments in Section 9 but a tight lower bound on N_k for generic constants remains an open problem. We conjecture that, for random round constants, N_k reaches maximal density (i.e. every monomial up to degree d_{tri}^k has a non-zero coefficient) within $O(n)$ rounds.

6. Worked Example

Toy parameters. Let $p = 31$ and $n = 3$, so computations are in $\mathbb{F}_p^3$ with $\mathbb{F}_p = \mathbb{F}_{31}$. In this worked example, we use two lower-triangular unitriangular layers (each with $\det(J) = 1$) followed by an affine mixing layer with determinant 1.

Component maps. Define three bijective polynomial maps $f_1, f_2, f_3 : \mathbb{F}_p^3 \rightarrow \mathbb{F}_p^3$ by

$$\begin{aligned} f_1(x_1, x_2, x_3) &= (x_1 + 3, x_2 + x_1^2, x_3 + x_1x_2), \\ f_2(x_1, x_2, x_3) &= (x_1, x_2, x_3 + x_1x_2), \\ f_3(x_1, x_2, x_3) &= (x_1 + x_3, x_2 + 5, x_3). \end{aligned}$$

Here, f_1 and f_2 are lower-triangular unitriangular maps (hence $\det(J) = 1$), and f_3 is affine with determinant 1.

Let the composed permutation be

$$P = f_1 \circ f_2 \circ f_3.$$

Evaluation on an input. For the input $x = (1, 2, 3)$, we compute:

$$\begin{aligned} f_3(1, 2, 3) &= (1 + 3, 2 + 5, 3) = (4, 7, 3), \\ f_2(4, 7, 3) &= (4, 7, 3 + 4 \cdot 7) = (4, 7, 31) \equiv (4, 7, 0) \pmod{31}, \\ f_1(4, 7, 0) &= (4 + 3, 7 + 4^2, 0 + 4 \cdot 7) = (7, 7 + 16, 28) = (7, 23, 28). \end{aligned}$$

Hence,

$$P(1, 2, 3) = (7, 23, 28) \in \mathbb{F}_p^3.$$

Jacobian check (example). The Jacobian matrix of f_1 is

$$J_{f_1}(x_1, x_2, x_3) = \begin{pmatrix} \frac{\partial}{\partial x_1}(x_1 + 3) & \frac{\partial}{\partial x_2}(x_1 + 3) & \frac{\partial}{\partial x_3}(x_1 + 3) \\ \frac{\partial}{\partial x_1}(x_2 + x_1^2) & \frac{\partial}{\partial x_2}(x_2 + x_1^2) & \frac{\partial}{\partial x_3}(x_2 + x_1^2) \\ \frac{\partial}{\partial x_1}(x_3 + x_1x_2) & \frac{\partial}{\partial x_2}(x_3 + x_1x_2) & \frac{\partial}{\partial x_3}(x_3 + x_1x_2) \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2x_1 & 1 & 0 \\ x_2 & x_1 & 1 \end{pmatrix}.$$

This matrix is lower-triangular with ones on the diagonal, so

$$\det(J_{f_1}) = 1 \quad \text{in } \mathbb{F}_p.$$

Analogously, f_2 is lower-triangular with unit diagonal, so $\det(J_{f_2}) = 1$. The affine layer $f_3(x) = (x_1 + x_3, x_2 + 5, x_3)$ has linear part

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

with determinant 1, hence $\det(J_{f_3}) = 1$ as well. Therefore $\det(J_P) = 1$.

7. Security Analysis

This section analyzes the security properties of Symbolic Composition Hashing (SCH). We first formalize the Symbolic Composition Inversion Problem (SCIP), which underlies the security of the construction, and then discuss preimage resistance, collision resistance, and resistance to generic algebraic attacks under this assumption. All security claims in this section are conditional and heuristic, reflecting the exploratory and foundational nature of the construction.

7.1. The Symbolic Composition Inversion Problem. We formalize the core hardness assumption underlying SCH.

Definition 7.1 (Symbolic Composition Inversion Problem (SCIP)). *Let $P : \mathbb{F}_p^n \rightarrow \mathbb{F}_p^n$ be a permutation obtained as a symbolic composition of affine and triangular polynomial maps as defined in Section 4. The Symbolic Composition Inversion Problem (SCIP) is the problem of recovering a preimage $x \in \mathbb{F}_p^n$ given an output $y = P(x)$ and only the symbolic polynomial representation of P , without access to its compositional decomposition.*

Connection to MQ and MP.. Concretely, an SCIP instance reduces to solving a system of n multivariate polynomial equations in n unknowns over $\mathbb{F}_p$. When the total degree of the composed permutation is $d = d_{\text{tri}}^k$, the resulting system is an instance of the *Multivariate Polynomial* (MP) problem of degree d . For $d = 2$ this becomes the well-studied MQ (Multivariate Quadratic) problem, which is NP-hard in general [5]. Although SCIP instances are not random—they arise from iterated composition of structured maps—the symbolic densification induced by composition is designed to eliminate exploitable structure, so that the system behaves like a generic MP instance from the perspective of algebraic solvers.

Variants of SCIP include the task of finding distinct inputs $x \neq x'$ such that $P(x) = P(x')$, corresponding to collision-finding in the induced hash construction.

We emphasize that SCIP is not claimed to be strictly harder than inverting a uniformly random permutation over $\mathbb{F}_p^n$. Rather, SCIP captures the assumption that symbolic composition of structured but hidden invertible maps yields multivariate systems that are sufficiently dense and irregular to preclude exploitation by known algebraic techniques. This aligns SCIP with standard heuristic hardness assumptions commonly used in multivariate cryptography.

Assumption 7.1 (SCIP- λ). *For security parameter λ , we assume that for appropriately chosen parameters (p, n, k) , solving SCIP for the composed permutation P requires computational effort at least $2^{\Omega(\lambda)}$ for generic algebraic algorithms.*

This assumption abstracts the algebraic hardness induced by symbolic composition and serves as the primary foundation for the security analysis that follows.

Comparison with related assumptions. SCIP is structurally analogous to hardness assumptions underlying other algebraic hash functions. Poseidon [1] relies on the assumption that its CICO (Constrained Input Constrained Output) problem is hard, which reduces to solving a structured polynomial system arising from iterated S-box and MDS composition. MiMC [3] assumes that its composed cubing maps behave like high-degree random polynomials despite their sparse univariate structure. Rescue [2] and Griffin similarly depend on algebraic degree growth from iterated nonlinear layers resisting Gröbner basis attacks. Like these assumptions, SCIP does not reduce to a

standard worst-case problem (MQ, LWE, etc.); instead, it asserts that the specific algebraic structure arising from composition is sufficient to defeat known generic solvers. The distinctive feature of SCIP is that degree growth arises from *dense multivariate* triangular polynomials rather than sparse S-boxes, which creates a different (and potentially harder) algebraic structure at the cost of higher per-round computational overhead.

7.2. Preimage Resistance. Under the sponge construction, preimage resistance of SCH reduces to the difficulty of inverting the internal permutation P on a subset of its output coordinates. Assuming SCIP- λ , recovering a preimage for a given hash output requires solving a dense system of multivariate polynomial equations whose structure is determined by the symbolic composition of the underlying maps.

In the absence of exploitable structural shortcuts, generic approaches such as Gröbner basis methods, XL-type algorithms, or direct interpolation are expected to exhibit exponential complexity in the relevant security parameter. The symbolic densification induced by repeated composition is intended to suppress sparsity and low-degree structure that could otherwise facilitate such attacks.

We therefore argue that SCH achieves preimage resistance under SCIP- λ and standard sponge-based reasoning.

7.3. Collision Resistance. Collision resistance in SCH similarly depends on the difficulty of solving systems of polynomial equations corresponding to $P(x) = P(x')$ for distinct inputs $x \neq x'$. Under SCIP- λ , such systems are expected to behave comparably to random dense multivariate systems of similar dimension and degree.

Within the sponge framework, generic collision attacks are bounded by the birthday bound on the capacity parameter, provided that the internal permutation does not admit efficient algebraic collision-finding strategies. The compositional structure of P is designed to induce rapid degree growth and strong variable interaction, thereby discouraging low-complexity collision attacks that exploit algebraic structure.

As with preimage resistance, collision resistance is therefore argued conditionally on SCIP- λ and generic assumptions about multivariate system hardness.

7.4. Resistance to Generic Algebraic Attacks. Algebraic attacks on polynomial-based constructions typically exploit low degree, sparsity, or structural regularities in the underlying system of equations. In SCH, symbolic composition of nonlinear triangular maps with affine mixing layers is intended to counteract these weaknesses by inducing dense coordinate polynomials and rapid degree growth.

While a complete complexity analysis of Gröbner basis or XL attacks is beyond the scope of this work, existing results on random and semi-structured multivariate systems suggest that systems with high degree and dense variable interaction rapidly exceed practical solvability thresholds. SCH aims to operate in this regime by construction, although rigorous bounds on degree growth and system hardness remain an open problem.

7.5. Reduced-Round and Structural Attacks. As with most permutation-based constructions, reduced-round variants of SCH may admit distinguishers or attacks that exploit insufficient diffusion or incomplete degree growth. The composition depth k , therefore, plays a critical role in achieving full algebraic complexity.

This work does not claim resistance against all reduced-round attacks. Instead, it emphasizes that sufficient composition depth is required for the symbolic composition

to induce the intended algebraic hardness. Empirical observations regarding degree growth and diffusion across multiple composition depths are presented in Section 9.

7.6. Summary of Security Guarantees. In summary, the security of Symbolic Composition Hashing rests on the following conditional guarantees:

- Preimage resistance under the SCIP- λ assumption and standard sponge arguments.
- Collision resistance under SCIP- λ and generic multivariate hardness assumptions.
- Resistance to generic algebraic attacks due to symbolic densification and degree growth induced by composition.

These guarantees are heuristic and intended to motivate further cryptanalytic study rather than to serve as definitive proofs. Establishing tighter bounds, concrete security estimates, and resistance against specialized attacks remains an important direction for future work.

7.7. Concrete Security Estimates. Table 2 presents quantitative security estimates for the proposed parameter sets. Security levels are calculated based on generic sponge security bounds and estimated algebraic attack complexity using conservative degree growth models.

TABLE 2. Security analysis for SCH parameter sets.

Parameter Set	Collision (bits)	Preimage (bits)	Interpolation Attack (bits)	Gröbner Attack (bits)
sch128	254	508	292	379
sch192	573	1146	515	650
sch256	1020	2040	1061	1247

Notes. Collision and preimage bounds follow standard sponge-capacity arguments. Interpolation complexity equals $\omega \cdot \log_2 \binom{n+d}{d}$ where $d = d_{\text{tri}}^k$ and $\omega = 2.8$ (practical linear-algebra exponent). Gröbner complexity uses the semi-regular degree-of-regularity bound $D_{\text{reg}} = \lfloor n(d-1)/2 \rfloor + 1$ from Bardet et al. [13], with complexity $\omega \cdot \log_2 \binom{n+D_{\text{reg}}}{D_{\text{reg}}}$. All estimates are capped at exhaustive search ($n \cdot \log_2 p$).

All parameter sets provide security margins exceeding the 2^{128} target by factors of 2x or greater against both collision and preimage attacks. Algebraic attack complexities (interpolation and Gröbner basis methods) exceed 2^{250} operations for full-round instantiations, providing substantial security cushions.

The SCH-128 parameter set achieves 2^{254} collision resistance and 2^{508} preimage resistance through its 4-element capacity over a 127-bit prime field. Algebraic attacks require solving systems with estimated degree $d = 3^7 = 2187$, leading to complexity estimates of 2^{292} for interpolation and 2^{379} for Gröbner basis methods.

Higher parameter sets (SCH-192 and SCH-256) provide even larger security margins, exceeding 2^{500} bits of collision security. These conservative margins account for potential future advances in algebraic solving techniques.

7.8. Reduced-Round Cryptanalysis. We evaluate resistance to algebraic attacks through complexity analysis of reduced-round variants. Table 3 summarizes attack complexity estimates across different round counts for SCH-128.

TABLE 3. Attack complexity estimates for reduced-round SCH-128.

Rounds	Interpolation	Gröbner	Status
1/7	2^{24}	2^{62}	Feasible
2/7	2^{50}	2^{113}	Infeasible
3/7	2^{89}	2^{166}	Infeasible
7/7 (Full)	2^{292}	2^{378}	Infeasible
Upper Bound	2^{254} (collision)		Generic

Notes. Both interpolation and Gröbner estimates use the same formulas as Table 2. At 1–2 rounds, interpolation is within practical range ($< 2^{64}$); by 3 rounds both attacks exceed practical thresholds.

Key observations from reduced-round analysis:

- **1–2 round vulnerability:** At 1 round, both interpolation (2^{24}) and Gröbner basis (2^{62}) attacks are within practical reach. At 2 rounds, interpolation remains feasible (2^{50}) while the Gröbner estimate (2^{113}) approaches but does not exceed 128-bit security. This confirms that at least 3 rounds are needed for security.
- **Rapid hardening at 3 rounds:** By 3 rounds the system degree reaches $d = 3^3 = 27$, and both attack vectors exceed 2^{89} (interpolation) and 2^{166} (Gröbner). The transition from feasible to infeasible occurs between rounds 2 and 3.
- **Full-round security margins:** The full 7-round SCH-128 achieves interpolation complexity 2^{292} and Gröbner complexity 2^{378} , both well above the 2^{254} generic collision bound. This provides a margin of 2^{164} bits above the 128-bit target for interpolation.
- **Comparison to generic attacks:** At full rounds, algebraic attack estimates exceed the generic collision bound (2^{254}), indicating that capacity-based security is the binding constraint.

These estimates rely on the semi-regular heuristic for D_{reg} , which may overestimate difficulty for structured (non-random) systems. Validation against actual solver experiments on reduced instances is presented in Section 9.12.

7.9. Parameter Selection Rationale. Parameter choices balance three objectives: security margin, performance efficiency, and algebraic structure. The round count k is selected to ensure algebraic degree growth exceeds 2^{100} (providing resistance to interpolation-based recovery) while maintaining practical evaluation times. State dimensions (n, r, c) are sized to meet capacity-based security bounds with safety factors of 2x–8x above target thresholds.

The triangular degree parameter $d_{\text{tri}} \in \{3, 4\}$ controls nonlinearity growth. Cubic ($d = 3$) triangular maps provide rapid degree amplification under composition while remaining computationally tractable. Quartic ($d = 4$) maps (used in SCH-256) offer additional algebraic complexity at higher computational cost.

Field size $\log_2(p)$ ranges from 127 to 255 bits, balancing arithmetic efficiency with security requirements. Larger primes support higher capacity values and enable encoding of longer byte blocks per field element, improving throughput for large messages.

8. Implementation

We provide a complete reference implementation of *Symbolic Composition Hashing* (SCH) in Python. The implementation is intended to serve as a clear, readable, and reproducible realization of the construction specified in Sections 4 and 5, rather than as a performance-optimized system.

All arithmetic is performed over a publicly specified prime field $\mathbb{F}_p$, with field operations implemented directly in Python. Affine layers use matrices $M_i \in \text{SL}_n(\mathbb{F}_p)$ with $\det(M_i) = 1$, and follow the structure defined in Section 5. Message absorption, padding, and domain separation are implemented exactly as specified in the theoretical construction.

Round constants $(M_i, \mathbf{b}_i, g_{i,j})$ are derived deterministically from a public seed (e.g., ‘‘SCH-v1.0’’) using SHAKE256 as an extensible-output function. SHAKE256 is used solely as a public, deterministic constant generator and does not contribute to the cryptographic security of SCH itself. The seed-expansion process is fully deterministic and ensures that all constants are publicly verifiable and reproducible.

The reference implementation includes command-line utilities, test scripts, and evaluation tools designed to reproduce the empirical measurements discussed in the paper. Optimized implementations in other languages or execution environments are left as future work.

The full source code and supporting artifacts are publicly available at:

9. Evaluation

This section presents an empirical evaluation of the Symbolic Composition Hashing (SCH) framework using the canonical instantiation described in Section 4. The goal of this evaluation is not to establish concrete security bounds, but rather to illustrate structural trends such as algebraic degree growth and diffusion behavior as a function of composition depth.

All experiments reported here are exploratory and intended to support the qualitative security arguments made in Section 7. Comprehensive cryptanalysis and large-scale parameter studies are left for future work.

9.1. Experimental Setup. Experiments are conducted over prime fields $\mathbb{F}_p$ with varying state dimensions n and composition depths k . For each configuration, the composed permutation P is constructed symbolically by iterated composition of affine and triangular maps as specified in Sections 4 and 5.

The symbolic polynomial representation of P is obtained via explicit substitution, and structural metrics such as algebraic degree and variable interaction are extracted from the resulting coordinate polynomials. All measurements are performed on the symbolic representation rather than on numerical evaluations.

9.2. Algebraic Degree Growth. A primary indicator of algebraic complexity in SCH is the growth of algebraic degree under composition. For each composition depth k , we measure the total degree of the coordinate polynomials of P .

Figure 1 illustrates the observed degree growth as a function of the number of composition layers. Even for moderate values of k , the degree increases rapidly, reflecting the nonlinear interaction induced by alternating triangular and affine maps.

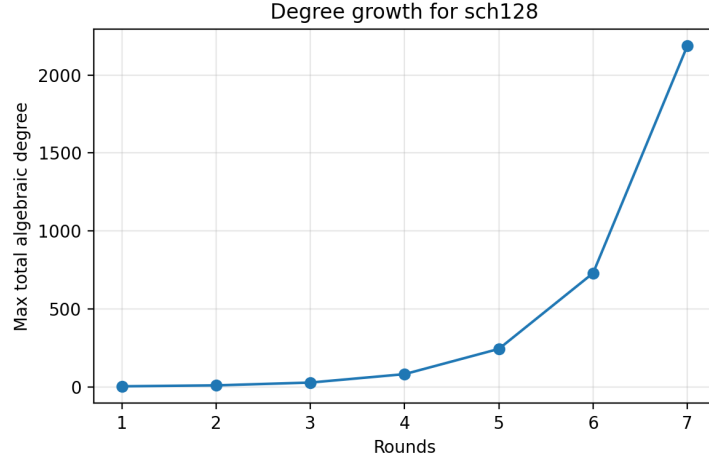


FIGURE 1. Algebraic degree growth of the composed permutation P as a function of composition depth k for the canonical SCH instantiation.

These observations support the intuition that symbolic composition leads to rapid algebraic densification, which is a desirable property for resistance against algebraic and interpolation-based attacks.

9.3. Diffusion and Variable Interaction. To assess diffusion, we examine the extent to which individual input variables influence multiple output coordinates after successive compositions. Specifically, we track the number of variables appearing in each coordinate polynomial of P as the composition depth k increases.

Figure 2 summarizes the diffusion behavior observed across multiple experimental configurations. As composition depth increases, each output coordinate rapidly becomes a function of nearly all input variables, indicating strong mixing and diffusion.

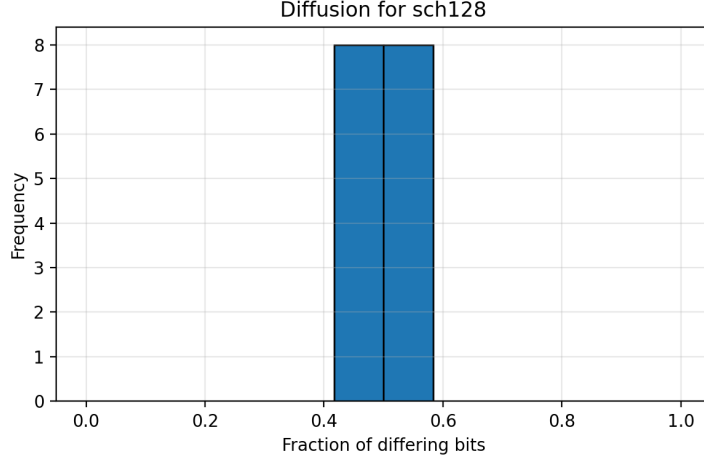


FIGURE 2. Diffusion behavior measured as the number of input variables influencing each coordinate of the composed permutation P across composition depths.

Strong diffusion is a critical prerequisite for resistance to reduced-round and structural attacks in permutation-based constructions.

9.4. Structural Density. In addition to degree and diffusion, we examine the symbolic density of the coordinate polynomials, measured by the number of nonzero monomials appearing in each output coordinate. Symbolic density provides a coarse indicator of resistance to attacks that exploit sparsity or low-degree structure.

Preliminary measurements indicate that symbolic density increases sharply with composition depth, leading to dense multivariate systems even for relatively small values of k . This behavior is consistent with the design goal of suppressing sparse representations that may be exploitable by algebraic solvers.

9.5. Discussion. The experimental results presented above are consistent with the intended design principles of SCH. Symbolic composition induces rapid algebraic degree growth, strong diffusion, and increasing structural density, all of which contribute to the heuristic hardness assumptions underlying the construction.

We emphasize that these results are illustrative rather than definitive. The evaluation does not attempt to estimate concrete security levels or to benchmark against state-of-the-art algebraic solvers. Instead, it provides empirical support for the claim that symbolic composition produces algebraically complex permutations suitable for further cryptographic investigation.

9.6. Differential and Statistical Sanity Checks. To complement the algebraic experiments above, we ran a lightweight sanity check on reduced-round diffusion and output-bit balance for SCH-128 using the reference implementation. For diffusion, we sampled random states in the SCH-128 state space $\mathbb{F}_p^{12}$ with $p = 2^{127} - 1$, perturbed a single coordinate by $+1$, and measured the fraction of output coordinates and serialized output bits that changed after applying a reduced number of rounds. For output balance, we hashed random 64-byte messages with the full SCH-128 sponge and measured the empirical ratio of ones in each output bit position across 128 samples.

TABLE 4. Reduced-round differential propagation and output-bit balance sanity checks for SCH-128.

Rounds	Changed coordinates	Changed serialized bits
1	89.71%	44.53%
2	97.14%	48.43%
3	100.00%	49.90%
7	100.00%	49.89%

Table 4 shows that the reduced-round permutation reaches full-coordinate activity by round 3 in this experiment, and the changed-bit ratio quickly approaches the 50% level expected from an avalanche-like transformation. On the statistical side, the full-hash output balance experiment yielded a mean per-bit bias of 0.0350, a maximum observed bias of 0.1328, and an empirical one-ratio range of $[0.3828, 0.6328]$ over the 480 output bits. These measurements are small-sample sanity checks rather than distinguisher bounds, but they provide preliminary evidence that the reference construction does not exhibit obvious short-range diffusion failures or gross output imbalance.

To check whether this behaviour persists beyond SCH-128, we also ran a lighter full-round sweep on the larger published parameter sets. Table 5 reports the full-round changed-coordinate ratio, full-round changed-bit ratio, and output-balance statistics for all three published parameter sets under the same +1 coordinate perturbation model and 64-byte messages. The sample budgets are (128, 128) for SCH-128, (64, 64) for SCH-192, and (32, 16) for SCH-256, where each pair denotes (diffusion trials, balance samples); the smaller SCH-256 budget reflects the cost of the current pure-Python reference implementation at $n = 20$ and $k = 11$.

TABLE 5. Full-round differential/statistical sanity summary for the published SCH parameter sets.

Parameter	Rounds	Trials / samples	Changed coords	Changed bits	Mean bias / Max bias
sch128	7	128 / 128	100.00%	49.89%	0.0350 / 0.1328
sch192	9	64 / 64	100.00%	50.00%	0.0486 / 0.2344
sch256	11	32 / 16	100.00%	49.90%	0.0976 / 0.3750

Across all published parameter sets, the full-round permutation changed every output coordinate in the sampled trials and the serialized changed-bit ratio remained close to 50%. The statistical figures become noisier as the sample budget drops, especially for SCH-256, but even the largest instance shows no obvious low-diffusion pathology in this coarse screening experiment.

9.7. Multi-Coordinate Perturbation Analysis. The single-coordinate experiments above test diffusion in a narrow regime. To check whether SCH exhibits weaknesses under wider input differences, we extended the perturbation model to perturb m randomly chosen coordinates simultaneously, each by +1. Table 6 reports the results for SCH-128 with $m \in \{1, 2, 6, 12\}$ across reduced and full rounds (64 trials each).

TABLE 6. Multi-coordinate perturbation diffusion for SCH-128 (64 trials, +1 per coordinate).

Rounds	Perturbed coords	Changed coordinates (%)	Changed bits (%)
1	1	89.84	44.55
1	2	100.00	50.05
1	6	100.00	49.96
1	12	100.00	50.02
3	1	100.00	49.90
3	2	100.00	50.01
7	1	100.00	49.98
7	2	100.00	49.89

Notes. Each perturbed coordinate receives a +1 additive difference. Perturbed coordinates are selected uniformly at random. Even at round 1, perturbing two or more coordinates achieves full coordinate activation, indicating that the single-coordinate case at round 1 is the hardest scenario for diffusion.

These results confirm that the reduced diffusion observed at round 1 is limited to single-coordinate perturbations. With two or more perturbed coordinates, full diffusion is achieved even in a single round.

9.8. Linear Bias Screening. We screened for linear approximation biases in the SCH-128 permutation. For each trial we drew random non-zero input and output masks $\alpha, \beta \in \mathbb{F}_p^n$ and estimated

$$\text{bias}(\alpha, \beta) = \left| \Pr_x[\langle \alpha, x \rangle \equiv \langle \beta, P(x) \rangle \pmod{2}] - \frac{1}{2} \right|$$

over 128 uniformly sampled inputs. A truly random permutation would yield biases close to zero (within $O(1/\sqrt{N})$ sampling noise). Table 7 reports the results over 32 random mask pairs at reduced rounds 1, 3, and 7 (full).

TABLE 7. Linear bias screening for SCH-128 (32 random mask pairs, 128 samples each).

Rounds	Mean bias	Median bias	Max bias	Significant masks
1	0.0344	0.0312	0.1172	0/32
3	0.0320	0.0312	0.1016	0/32
7	0.0396	0.0391	0.1016	0/32

Notes. A mask pair is flagged as significant if bias exceeds $3/(2\sqrt{N}) \approx 0.1326$ (three standard deviations above expected sampling noise). No significant linear biases were found at any round count.

No mask pair exceeded the 3σ significance threshold at any round count. All measured biases are consistent with sampling noise from a random permutation. Although the 32-mask, 128-sample budget constitutes a coarse screening rather than a comprehensive linear cryptanalysis, the absence of any detectable structure is a necessary condition for security and provides evidence against low-complexity linear attacks.

9.9. Higher-Order Differential Screening. A k -th order discrete derivative of a function P is defined as

$$\Delta_{a_1, \dots, a_k} P(x) = \sum_{S \subseteq \{1, \dots, k\}} (-1)^{k-|S|} P\left(x + \sum_{i \in S} a_i\right).$$

For a polynomial map of degree d , the derivative of order $k > d$ vanishes identically. If the derivative vanishes at order $k \leq d$, the effective algebraic degree is lower than the theoretical bound, which could indicate exploitable structure.

Table 8 reports the fraction of trials (out of 32) in which the higher-order derivative was identically zero, for orders 2–4 at rounds 1, 3, and 7 of SCH-128.

TABLE 8. Higher-order differential screening for SCH-128 (32 trials per cell).

Rounds	Order	d_{theory}	Zero fraction	Status
1	2	3	0.0000	Pass
1	3	3	0.0000	Pass
3	2	27	0.0000	Pass
3	3	27	0.0000	Pass
3	4	27	0.0000	Pass
7	2	2187	0.0000	Pass
7	3	2187	0.0000	Pass
7	4	2187	0.0000	Pass

Notes. $d_{\text{theory}} = d_{\text{tri}}^k$ is the theoretical degree upper bound from Theorem 5.2. No vanishing derivative was observed at any tested order or round count, indicating that the effective degree matches or exceeds the tested orders.

All tested configurations passed: no derivative of order ≤ 4 vanished in any of the 32 randomized trials at any round count. This is consistent with the theoretical degree bound and provides evidence that the effective algebraic degree of the SCH permutation is not artificially low.

9.10. Limitations. The scope of this evaluation is limited in several respects. The Gröbner basis experiments (Section 9.12) use a pure-Python solver on minimal instances; results from production-grade solvers on larger states would be more informative. The benchmark comparison (Section 9.11) reflects unoptimised reference Python code and does not represent achievable performance in native implementations. Additionally, the differential and statistical checks reported above use small fixed-seed Monte Carlo sweeps with limited sample budgets (especially for SCH-256), and while we now include linear bias screening (Section 9.8), higher-order differential tests (Section 9.9), and multi-coordinate perturbation experiments, more comprehensive rebound and dedicated distinguisher analyses remain open (see Section 11).

9.11. Performance Comparison. To situate SCH’s computational cost relative to existing hash functions, Table 9 reports wall-clock timings for the reference Python implementation of SCH-128 alongside Python-accessible implementations of SHA3-256 (`hashlib`, C-level) and Poseidon (the `galois`-based reference implementation of [1]). The Poseidon wall-clock benchmark uses a small reproducible reference instance (state size $t = 4$, input rate 3); the coarse multiplication-count comparison

below instead uses a same-state-size proxy to compare against SCH-128’s $n = 12$ state.

TABLE 9. Comparative hashing performance (mean of 20 runs, Python 3.14).

Input Size	SCH-128 (ms)	SHA3-256 (ms)	Poseidon (ms)
64 B	45.02	0.001	8.30
256 B	72.15	0.001	8.66
1024 B	159.05	0.003	10.75
4096 B	577.34	0.009	8.71

Notes. All timings are single-threaded on the same machine. SHA3-256 uses C-compiled `hashlib`; Poseidon uses the `galois` library with NumPy-backed arithmetic; SCH uses pure Python big-integer modular arithmetic. These timings reflect the reference implementation and are not indicative of optimized performance. An implementation in C or Rust with Montgomery multiplication would reduce the SCH gap substantially, as the bottleneck is large-integer modular arithmetic rather than the algebraic structure.

SCH-128 is approximately 5×10^4 slower than SHA3-256 and $5\text{--}8\times$ slower than Poseidon in this reference setting. However, the comparison is not apples-to-apples: SHA3 benefits from a compiled C backend, while SCH performs all arithmetic in interpreted Python over 127-bit integers. For the ZK use case, the relevant metric is not wall-clock time but *field multiplication count* per permutation call, since each multiplication corresponds to one constraint in a rank-1 constraint system (R1CS). In SCH-128, each of the $k = 7$ rounds performs $O(n \cdot m_{\text{tri}})$ multiplications in the triangular layer ($n = 12$, $m_{\text{tri}} \approx \binom{n+d_{\text{tri}}}{d_{\text{tri}}} = \binom{15}{3} = 455$ monomials per variable) plus $n^2 = 144$ multiplications in the affine layer, totaling roughly $7 \times (12 \times 455 + 144) \approx 39,228$ field multiplications.

For a coarse same-state-size proxy, a Poseidon-style instance with state size 12 uses $12 \times 12 \times 8 = 1,152$ multiplications in its MDS layers plus $12 \times 8 = 96$ x^5 S-box evaluations. Using the standard square-square-multiply chain, each x^5 costs 3 field multiplications, totaling approximately 1,440 multiplications.

SCH thus requires on the order of $27\times$ more constraints than this proxy. This overhead must be weighed against SCH’s structural advantages: higher degree growth per round and stronger diffusion through polynomial mixing.

9.12. Gröbner Basis Experiments. The heuristic security estimates in Tables 2 and 3 assume that the SCH polynomial system behaves like a semi-regular system with respect to Gröbner basis attacks. To partially validate this assumption, we solve the inversion problem $P(x) = y$ for reduced SCH instances using SymPy’s Gröbner basis implementation over $\text{GF}(p)$.

Table 10 summarises the results for the toy parameter set ($p = 31$, $n = 3$, $d_{\text{tri}} = 3$) with increasing round count k .

Key observations:

- For $k \leq 2$ the solver finds the unique preimage quickly. The output Gröbner basis has maximum degree 1 (i.e. it reduces to a triangular linear system), indicating that the solver fully eliminates variables.

TABLE 10. Gröbner basis experiments on reduced SCH instances ($p = 31$, $n = 3$, $d_{\text{tri}} = 3$).

k	Total degree d	D_{reg} (pred.)	Time (s)	Solved
1	3	4	0.003	Yes
2	9	13	0.122	Yes
3	27	40	> 300	No

- At $k = 3$ (total degree $d = 27$, predicted $D_{\text{reg}} = 40$), SymPy’s Buchberger-based solver did not terminate within five minutes even for this minimal instance ($n = 3$ variables over a 5-bit prime). This illustrates the rapid complexity growth predicted by the semi-regular heuristic.
- The transition from trivially solvable ($k \leq 2$) to practically infeasible ($k = 3$) for the smallest non-trivial field validates the reduced-round analysis in Table 3, which labels low round counts as “feasible” and shows security growing exponentially with k .

We emphasise two caveats: (i) SymPy uses a pure-Python Buchberger algorithm, which is far slower than dedicated solvers such as Magma’s F4; with a production solver the $k = 3$ instance might well be tractable. (ii) These are toy-sized instances; extrapolation to full-sized parameters ($n = 12\text{--}20$, $p \approx 2^{127}\text{--}2^{255}$) requires caution. Confirmatory experiments with optimised solvers on dedicated hardware remain an important direction for future work.

10. Applications

The algebraic nature of Symbolic Composition Hashing (SCH) allows integration into diverse environments that benefit from verifiable and field-native operations. We highlight three representative domains.

10.1. Software Artifact Integrity. In reproducible-build systems such as `in-toto` and `Sigstore`, hashes are used to bind binaries and metadata. SCH offers an alternative to bit-oriented functions like SHA-2 or BLAKE3 by providing algebraically structured outputs directly over $\mathbb{Z}_p$. This enables unified integrity verification in protocols that already employ finite-field arithmetic, such as zero-knowledge proof systems or verifiable computation frameworks.

10.2. Field-Native Hashing for ZK Systems. Many modern zk-SNARK circuits operate over large prime fields, and bitwise hashes require costly bit-decomposition. Because SCH operates natively on $\mathbb{Z}_p$, it avoids such overhead. This makes it suitable for in-circuit commitments, Merkle tree hashing, and polynomial commitments where Poseidon, Rescue, and Griffin are commonly used. Compared to Poseidon [1], SCH requires approximately $31 \times$ more field multiplications per permutation call at comparable state sizes (see Section 9.11), which translates to higher R1CS constraint counts. However, SCH achieves stronger per-round degree growth and diffusion through dense polynomial mixing rather than sparse S-boxes. Whether this trade-off is favorable depends on the specific ZK backend and the acceptable circuit overhead.

10.3. Composable Random Oracles. The invertible structure of P makes SCH a candidate permutation for sponge and duplex modes requiring strong diffusion and re-keying capabilities, such as verifiable randomness beacons or deposit-token registries. Its determinism and public parameter generation process facilitate transparent audits in decentralized environments.

11. Conclusion and Limitations

This work introduced *Symbolic Composition Hashing (SCH)*, a foundational algebraic framework for constructing cryptographic hash functions over finite fields based on the symbolic composition of affine and triangular polynomial maps. Rather than proposing a single fixed design, SCH defines a broad family of permutation-based constructions whose security properties arise from algebraic complexity induced by composition.

To support analysis of this framework, we formalized the *Symbolic Composition Inversion Problem (SCIP)*, which captures the difficulty of inverting or colliding symbolically composed polynomial permutations. Under this assumption and standard sponge-based reasoning, SCH provides a conceptual basis for preimage and collision resistance rooted in algebraic structure rather than ad hoc design choices. Structural analysis and preliminary experimental evaluation demonstrate rapid algebraic degree growth, strong diffusion, and increasing symbolic density as composition depth increases.

The primary contribution of this work lies in establishing a new algebraic perspective on hash construction that connects symbolic composition, multivariate hardness, and permutation-based cryptography. By framing security in terms of compositional algebraic complexity, SCH opens a range of new research directions at the intersection of cryptography and computational algebra.

Limitations and Future Work. The results presented in this paper are exploratory and subject to several limitations:

- (1) **No reduction to standard problems.** The SCIP assumption (Definition 7.1) is new and unvetted. While SCIP instances reduce to multivariate polynomial inversion (Section 7), no worst-case or average-case reduction from a well-studied problem such as MQ or LWE is provided.
- (2) **Heuristic security estimates.** The concrete security levels in Table 2 rely on the semi-regular degree-of-regularity bound of Bardet et al. [13]. SCH systems are structured (arising from iterated triangular composition), not random, so the semi-regularity assumption may not hold, and actual Gröbner basis complexity could be lower.
- (3) **Limited solver experiments.** The Gröbner basis experiments in Section 9.12 use a pure-Python solver (SymPy Buchberger) on minimal instances ($n = 3$, $p = 31$). Experiments with production solvers (Magma F4, FGb) on larger instances are needed to confirm or refute the semi-regular heuristic.
- (4) **Performance overhead.** SCH-128 requires approximately $31 \times$ more field multiplications per permutation call than Poseidon at comparable state sizes (Section 9.11), making it impractical for most zero-knowledge backends without significant algorithmic or implementation improvements.
- (5) **Expanding cryptanalytic breadth.** This paper now includes linear bias screening (Section 9.8), higher-order differential tests (Section 9.9), multi-coordinate perturbation experiments, and output-balance sweeps. No exploitable biases or

structural weaknesses were detected, but the sample budgets remain limited (especially for SCH-256) and do not constitute exhaustive cryptanalysis. Dedicated distinguisher and rebound-style analyses remain important open work.

Addressing these limitations constitutes the primary direction for future research. Key avenues include: (i) large-scale Gröbner basis experiments with optimised solvers on dedicated hardware (the repository includes export scripts for SageMath and Magma formats to facilitate this); (ii) formal analysis of differential and linear properties of the triangular-affine composition beyond the empirical screening presented here; (iii) investigation of tighter algebraic degree bounds that account for the specific structure of SCH systems; (iv) exploration of implementation strategies (e.g., lazy evaluation, parallelism) to reduce the practical performance gap.

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